Dropout#

https://pytorch.org/docs/stable/generated/torch.nn.modules.dropout.Dropout.h

Description:

During training, randomly zeroes some of the elements of the input tensor with probability p. The zeroed elements are chosen independently for each forward call and are sampled from a Bernoulli distribution. Each channel will be zeroed out independently on every forward call. This has proven to be an effective technique for regularization and preventing the co-adaptation of neurons as described in the paper Improving neural networks by preventing co-adaptation of feature detectors . Furthermore, the outputs are scaled by a factor of 11−p\frac{1}{1-p}1−p1 during training. This means that during evaluation the module simply computes an identity function.

Function Signature

```
class torch.nn.modules.dropout.Dropout(p=0.5, inplace=False)
[source]#

Parameters

Shape:

forward(input)[source]#

Return type
```

Returns:

Tensor

Code Examples

```
>>> m = nn.Dropout(p=0.2)
>>> input = torch.randn(20, 16)
>>> output = m(input)
```

Detailed Documentation

Documentation

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The zeroed elements are chosen independently for each forward call and are sampled from a Bernoulli distribution.

Each channel will be zeroed out independently on every forward call.

This has proven to be an effective technique for regularization and preventing the co-adaptation of neurons as described in the paper Improving neural networks by preventing co-adaptation of feature detectors .

Furthermore, the outputs are scaled by a factor of 11−p\frac{1}{1-p}1−p1 during training. This means that during evaluation the module simply computes an identity function.

p (float) – probability of an element to be zeroed. Default: 0.5

inplace (bool) – If set to True, will do this operation in-place. Default: False

Input: (∗)(*)(∗). Input can be of any shape

Output: (∗)(*)(∗). Output is of the same shape as input

Runs the forward pass.